



Monitoring path dependence in multi-technological organizations

Stefanie Bröring^a, Anna Waßenhoven^{a,*}, Silvan Berg^b, Michael Rennings^a

^a Ruhr University Bochum, Center for Entrepreneurship, Innovation and Transformation (CEIT), Chair for Entrepreneurship and innovative Business Models, Bochum, Germany

^b Lexis Nexis Intellectual Property Solutions, Bonn, Germany

ARTICLE INFO

Keywords:

Path dependence
Multi-technological organizations
Patent analysis
Oil and gas industry

ABSTRACT

In dynamic, social and technological environments, organizations must adopt more efficient technologies. For this to be possible, organizations need to maintain sufficient flexibility and, thus, need to be aware of potential technological path dependence. Technological path dependence occurs when an organization's past technological choices and developments create a narrowing corridor of possibilities, eventually locking the organization into specific patterns of technological behavior that are difficult to change. However, monitoring dependence on distinct technological paths is tremendously challenging. Literature largely neglects the question of when technological developments of organizations start to become path dependent and how this can be anticipated. By distinguishing between product-radical, process-radical, product-incremental, and process-incremental patents, as well as developing a path dependence indicator (PD_t), we provide empirical evidence of the emergence of technological path dependence in multi-technological organizations. Applying this indicator to the oil and gas sector reveals that lower PD_t values correspond to higher adoption rates of emerging technologies, which can lead to new technological paths. Moreover, our results support previous findings that emphasize the role of the surrounding industry in impacting path dependence.

1. Introduction

Technological change has always challenged incumbent organizations to renew their competence base. Extant research, however, has demonstrated that transition processes, more precisely the adoption of new technology alternatives and the involved organizational change, is hampered by path dependence [1–4]. As such, path dependence describes a process in which an organization's historical decisions increasingly limit its future possibilities and ultimately trap the organization in rigid, repetitive decision-making patterns. It illustrates how early investments in certain capabilities and infrastructure create self-reinforcing cycles that increase efficiency but inadvertently “lock-in” a company to existing technologies and limit its ability to innovate [5–7]. In fact, the concept of path dependence, which originates from economic theory [8,9], has witnessed a huge level of academic interest along different research lines. Scholars have, for example, focused on the effect of path dependence on new technology adoption, organizational learning and culture [10–12]. Additionally, extant literature discusses the concept of path dependence as an “antipode” to strategic renewal [13–15]. Thus, it is obvious that preventing a company

from getting trapped by its past and its extant knowledge corridor seems like a major task for any new business development or strategic planning executive [16–18]. This is even more challenging for multi-technological organizations which are built upon different technological paths and, thereby, operate in several distinguishable technology fields [19]. In contrast to organizations' well-studied, diverse technology portfolios, multi-technological organizations face the challenge of managing competing or even mutually excluding technologies in different paths of their portfolio [20].

However, in literature little attention has been paid to the question when organizations start to become path dependent from a technological perspective and if such a process can be anticipated to enable a timely action and thus prevents the pitfalls associated with the negative sides of path dependence [cf. 5,20]. This question seems increasingly relevant as it may help to handle technology dynamics potentially leading to a trapping lock-in situation limiting a corporation's ability to recognize opportunities outside its established technological paths. Management is often not aware of the need to act against a prospective situation of path dependence. This is because organizations trapped by their past success might be unaware of the need for strategic renewal,

* Corresponding author. Ruhr University Bochum, Center for Entrepreneurship, Innovation and Transformation (CEIT), Chair for Entrepreneurship and innovative Business Models, Suttner-Nobel Allee 4, 44803, Bochum, Germany.

E-mail address: anna.wassenhoven@ruhr-uni-bochum.de (A. Waßenhoven).

<https://doi.org/10.1016/j.wpi.2026.102455>

Received 26 May 2025; Received in revised form 23 March 2026; Accepted 23 March 2026

Available online 30 March 2026

0172-2190/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

which is even more relevant in cases of high environmental turbulence reflected by disruptive change and the emergence of novel technological regimes, such as in sustainability transitions [1,2,17,21,22].

Still, only few studies have attempted to empirically assess technological path dependence. Indeed, most studies in this research domain have applied a qualitative research design embedded in a particular research context [23,24]. Thus, literature would benefit from a quantitative approach that is able to classify the degree of technological path dependence in an organization's technology portfolio. Additionally, especially multi-technological organizations that strive to not only pursue different paths at the same time, would benefit from a monitoring approach to anticipate potential lock-in situations that emerge as a result of path dependence [20,25]. In any case, organizations need to be aware of emerging path dependencies over time if they wish to maintain strategic flexibility [4,26–28].

Consequently, this paper seeks to provide guidance by making implicit information and arbitrary assumptions of a likely “lock-in” situation explicit and measurable to enable actors to react in a timely manner. We provide empirical evidence for the emergence of technological path dependence in multi-technological organizations and apply our patent-based approach to the oil and gas industry – an industry that offers rich empirical data and includes actors that are witnessing a plethora of different emerging clean technologies challenging the current status quo [29–32]. More precisely, as this industry mainly relies on fossil resources it is faced with several challenges of sustainability transitions while lock-in mechanisms are one of the major hurdles to transform organizations towards a more sustainable economy [33].

We structure our paper as follows. In chapter 2, the study reviews the concept of path dependence and outlines how self-reinforcing mechanisms trigger the emergence of path dependence during technology development. Additionally, this chapter examines previous research of path-breaking approaches to highlight the importance of monitoring the emergence of path dependence for organizations, that is to maintain strategic flexibility along technological trajectories. In chapter 3, we introduce the path dependence indicator (PD_t) and apply it, in chapter 4.1. and 4.2., to the patent portfolio of Royal Dutch Shell as an exemplary case study. In chapter 4.3., these results are discussed and compared with PD_t values from other oil- and gas-producing organizations, that are Chevron Corporation, BP, and ExxonMobil. By also investigating companies' annual reports as well as their investments in clean technology (by means of patents as well as mergers and acquisitions data), we were able to further discuss and compare results of our analysis. In chapter 5, the strengths and weaknesses of the proposed methodological design are discussed in relation to existing approaches [e.g., 25,34–36], leading to managerial implications and directions for future research.

2. Path dependence in multi-technological organizations

The concept of path dependence has roots in evolutionary economics, where it is described as “an attempt to look at an economic system [...] as a continuing process in space and time” [9], or discussed as the central notion that “history matters” [37]. However, the concept of path dependence has also become increasingly relevant for organizational research as a way to explain the origins and evolution of technological regimes [6,38–41]. Here, the concept stresses how cumulating expertise and investments in scale efficiencies can act as self-reinforcing mechanisms that ultimately, though unintentionally, constrain the innovation capabilities of organizations by fostering the “lock-in” of prevailing technologies [6,7,25,39,42–44]. In this manner, Sydow et al. [6] describe three phases of path dependence, namely pre-formation, formation and lock-in (see Fig. 1). The pre-formation phase covers the randomly oriented research activities for a novel technological path. The formation phase is characterized by a narrowed spectrum of relevant research solutions for the given technological path and organizations strive to realize scale, scope, and learning effects for

the given technology. Finally, the lock-in phase describes how the repetitive decision-making process converges to a fixed point in which it locks-in itself.

When describing the trajectories of organizations or technologies, research has tended to use the concepts of path dependence and inertia rather interchangeably [37,45–47], although there are a number of important differences. While “trajectory” is generally considered as a by-process of organizational structures in the concept of inertia [48,49], the trajectory of path dependence refers to a particular path that is being followed during technology development, and one that may not necessarily apply to the whole organization [6,36,50,51]. For example, multi-technological organizations may simultaneously occupy and develop along several paths, each subject to its own degree of path dependence, however, here the capacity to be aware of path dependence is particularly challenging [20,35]. Co-existing technological paths present not only a conceptual dilemma but also a practical challenge for organizations. Given that the adoption of new technologies is a necessity for organizational survival [52], organizations need to be aware of path dependence [6,34,53–55]. However, managers are often unaware of path dependence issues and lack skepticisms about self-reinforcing mechanisms [27,53]. Further, examples of persistence and self-reinforcement can also be found across the surrounding industries in which organizations operate, specifically owing to the fact that similar decisions have been reproduced by many actors over time [34,35].

Multi-technological organizations do not diversify technology paths in a random manner but align them to those sectors in which historically developed knowledge and capabilities can be reused [56]. Thus, the concept of path dependence remains somewhat insufficient for the examination of multi-technological paths, given that such paths co-exist alongside one another, even while following their own dynamics [20]. Still, only few approaches target path dependence for multi-technological organizations in technology developments. In an extant literature review on the assessment of path dependence, we identified 2902 studies up to February 2023.¹ Out of these studies, around 1700 studies examine path dependence in the context of organizations, while only seven seek to measure path dependence in multi-technological organizations² (see Table 1).

Within these seven studies, several distinct methodological clusters emerge. Belussi and Sedita [57], Ibrus [23], and Klitkou et al. [24] all utilize qualitative meta-case study designs to explore evolutionary life cycles and micro-level organizational dynamics. However, these approaches often lack the broader transferability required for direct practical application. Similarly, Gómez and Atun [58] rely on qualitative content analysis of scientific articles, media and research reports to illustrate how coalition-building shapes governance in global health. In contrast, other scholars prioritize scalable, data-driven frameworks. The methodological approach of Bergek and Onufrey [20], and the subsequent work of Onufrey and Bergek [35], leverages publicly available data to investigate the self-reinforcing mechanisms of path dependence,

¹ Database: Web of Science. Search String: TS=(measur* OR analyz* OR analys* OR assess* OR calculat* OR empirical OR experimental) AND TS=(“path dependence” OR “path dependency”); Timespan: All years. Indexes: SCI-EXPANDED, SSCI, A&HCI, ESCI, 2902 results, February 09, 2023.

² Database: Web of Science; Search String: TS=(measur* OR analyz* OR analys* OR assess* OR calculat* OR empirical OR experimental) AND TS=(“path dependence” OR “path dependency”) AND TS=(organisation* OR organization* OR firm* OR institution* OR compan* OR enterprise* OR corporation* OR business*) AND TS=(“multi-technolog*” OR “multi-path” OR “multiple-technolog*” OR “multiple-path” OR “multi technolog*” OR “multi path” OR “multiple technolog*” OR “multiple path” OR “various technolog*” OR “various path” OR “several technolog*” OR “several path” OR “different technolog*” OR “different path” OR “variety technolog*” OR “variety path” OR “varying technolog*” OR “varying path”). Timespan: All years. Indexes: SCI-EXPANDED, SSCI, A&HCI, ESCI, 13 results (7 after full-text screening), February 09, 2023.

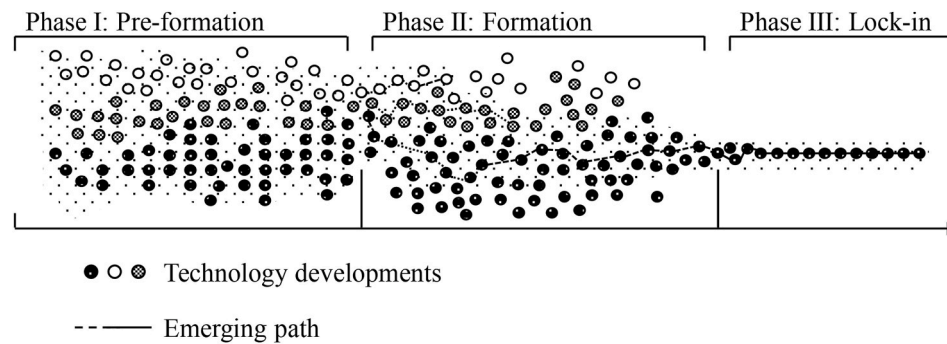


Fig. 1. The three-phases of path dependence. Source: Own illustration based on Sydow et al. [6].

Table 1

Literature review of approaches assess path dependence in multi-technological organizations.

Author	Title	Method and Data	Contribution
Belussi and Sedita [57]	Life Cycle vs. Multiple Path Dependency in Industrial Districts	Meta case study analysis on case studies of Italian industrial districts	Investigating the evolutionary process of industrial districts and its role of path dependence
Gómez and Atun [58]	Emergence of multilateral proto-institutions in global health and new approaches to governance	Content analysis of journal articles, books, agency web sites, media and research reports	Illustrating the differences in coalition formation processes that shape governance
Ibrus [23]	Path dependencies in media design: evolutionary dynamics of early mobile web and its textual forms	Meta case study analysis on early evolution of mobile web and its textual forms	Studying the dynamics that condition the evolution of various path dependencies of media systems
Klitkou et al. [24]	The role of lock-in mechanisms in transition processes: The case of energy for road transport	Meta case study analysis on advanced biofuels, e-mobility and hydrogen and fuel cell electrical vehicles	Highlighting how different path dependencies have been reinforced by the lock-in mechanisms
Onufrey and Bergek [35]	Self-reinforcing Mechanisms in a Multi-technology Industry	Building upon Bergek and Onufrey [20] that uses patent data of lightning industry	Investigating what kind of self-reinforcing mechanisms are present in path dependence industries
He and Walheer [59]	Spillovers and Path Dependences in the Chinese Manufacturing Industry: A Firm-Level Analysis	Three criteria partition of Chinese firm-level data assessed by measures of efficiency and technological advancement in data envelopment analysis-based methodology	Test presence of path dependences for firm efficiency and technological advancement
Malherbe and Simon [60]	Near-Field Communication (NFC) technology emergence: One or several technological paths?	Multilevel and longitudinal investigation of on-side observations, interviews, archival data and patent data	Understanding influence tactics on path stability vs. flexibility and dynamics of path constitution

though they primarily maintain a meso-perspective. Shifting to a more granular quantitative lens, He and Walheer [59] apply empirical analysis to Chinese firm-level data to test for effects of path dependence and spillover on efficiency and technological advancement in “mult-output

firms”. Finally, Malherbe and Simon [60] bridge these perspectives through a longitudinal, multi-level investigation of NFC technology, examining how specific influence tactics balance path stability with strategic flexibility. Despite their diverse insights, the studies lack a transferable, quantitative framework that can be practically applied to publicly available data to measure path dependence within multi-technological organizations.

Possibilities to overcome path dependence can be summarized as path transformation and dissolution mechanisms [26]. Such mechanisms are highly dependent upon the interaction between technology and power [61] and require leveraging effects such as recombining knowledge, motivation for participation and guaranteeing transparency [62]. The power for breaking old paths and creating new ones may be endogenously or exogenously driven [20]. Especially in corporate cultures with low emphasis on endogenous change, breaking path dependence requires external experts that can help to identify this pattern and develop capabilities to change certain routines [54]. Moreover, Bergek and Onufrey [20] show that new technological paths can evolve endogenously from existing structures. When shocks are driven endogenously, the technological path can interact across different domains of knowledge, which is particularly promising for creating novel technologies in multi-technological organizations [20].

3. Assessing path dependence in multi-technological organizations

The main objective of this paper is to analyze path dependence in distinctive technology developments of multi-technological organizations. Therefore, we use patent data as a proxy for technology developments in organizations. Patents contain a standardized structure, provide an accurate source for the analysis of cumulative technological developments and have been used to distinguish between multi-technical paths or different technological areas [20,35,63,64]. Each patent document contains bibliographic and textual information (e.g., abstract, title, applicant, citation and publication date) and, thus, patent data is a precise source for data and text mining techniques [63–66].

3.1. Oil and gas industry as a research site

We select the oil and gas industry as our research site as it reflects a sector with high environmental turbulence due to the needed energy transition and constitutes of a set of multi-technological organizations that bear a rich set of patents, since patenting is a key parameter for global competitiveness in this particular sector [67,68]. In recent decades, mergers and acquisitions have led to four organizations dominating the global oil and gas industry: Royal Dutch Shell, BP, Chevron Corporation and ExxonMobil. For many years, these multi-technological organizations have dominated the industry, maintaining vertical and horizontal integrated value chains within the industry [30,31]. More precisely, these organizations are active in upstream, midstream and downstream operation units [69]. In the upstream sector, the

organizations focus on the exploration and exploitation of crude oil and natural gas fields. In the midstream sector, the organizations manage the storage and transport of oil and gas, while in the downstream sector they handle the processing of oil and gas-based products including fuel refinery, gasification and chemicals [69]. Within the majority of these multiple-technological paths, various research has demonstrated trajectory and persistence [29–32]. Such lock-in effects may result from the interdependence of technological systems and governing institutions [70–73].

Deviating from existing technological paths, let alone the creation of new ones, represents a difficult undertaking for organizations in such mature industries [4,24,35]. Given the rapidly growing market potential and political support for renewable, bio-based products [74,75], this market represents a potential area to create new technological paths for oil and gas organizations. For example, the European Commission adopted new rules for biofuels and biogas as part of the Renewable Energy Directive, ensuring a minimum share of 14% of renewable energy in transport by 2030 [76].

3.2. Research method

Overall, the methodological approach includes three major steps (see Fig. 2). **Within step 1**, patent data was compiled for the selected organizations (including their subsidiaries) and the defined observation period. The analysis is based on INPADOC patent families. Consistent with prior research [e.g., 77,78], the unit of analysis is the patent family rather than individual patent documents. To avoid double counting the same invention across multiple jurisdictions, the earliest filed patent application within each INPADOC family was selected as the representative document. Only granted patents were included in the dataset. Patent families were excluded if the representative application lacked an English title, abstract, or application date. Within the next step, we use the textual content of the patents as basis to assign patents to different technological paths as well as to distinguish between product, process, incremental and radical characteristics. Therefore, in step 1, we extract the patents' title and abstract as both provide concentrated information about the patents' orientation [65].³

Within step 2, the methodological approach builds on two separate sub-steps. In the first sub-step, we identify technology-related keywords using the presented patent data to identify operation units as well as respective technological paths (for a list of keywords see Appendix Table A.1 and Table A.2). In fact, the approach calculates the most commonly used n-grams based on the textual content of the title and abstract of the selected patent data, similar to other frameworks proposed in literature [see e.g., 82,83]. Thereby, we consider different types of n-grams, that is unigrams (one-worded) and bigrams (two-worded), which have unique and distinct characteristics and can be used as technology-related keywords. Next, based on annual reports from the selected organization as well as consultation of an oil and gas industry expert, the derived technology-related keywords are clustered in accordance with the given operation units as well as the technological paths. It should be noted that patents are assigned explicitly to only one technological path according to the frequency of matching technology-related keywords and, thus, need to have an assignment level above a 60% threshold. For example, if Patent 1 matches two technology-related keywords of path Y and one technology-related keyword of path X, Patent 1 is assigned to path Y with an accuracy assignment level above 60% threshold. This 60% threshold level has been tested and approved on a sub-sample.

³ As the WIPO [79] states the abstract of a patent should "ascertain quickly the character of the subject matter covered by the technical disclosure [and] merely serves the purpose of technical information". Abstract and title of patent documents have been successfully used in studies with contextual patent analysis [80,81].

In the second sub-step, titles and abstracts are searched for specific keywords and associated prepositions that reflect process characteristics in patents, as well as specific keywords that reflect an incremental degree of innovation (see Table 2). A distinction between product and process, as well as radical and incremental adaptations is well suited to measure organizations innovativeness and persistence in technology development [84]. Product characteristics indicate that the innovation represents a rather new technology or a combination of new technologies, while process characteristics imply simultaneous changes to improve productivity in a process through specialization, increased material flows and standardized products [85]. However, the distinction between product and process characteristics for technology persistence is lacking in process industries such as the oil and gas, chemical or pulp and paper industries [86]. In the oil and gas industry, process innovations are closely connected to product innovations [87,88] which is why the oil and gas industry is also considered as a process-oriented industry [89]. This is because achieving quality standards for novel products inevitably requires process innovations [87,90]. Consequently, the introduced approach needs to further distinguish between radical and incremental properties in patent data. Examples how the specific keywords help to identify characteristics in patents are presented in Appendix (see Appendix Table A.3 and Table A.4). If a title and/or abstract of a patent contains no certain process (incremental) keywords, it is assumed to have product (radical) characteristics. Similar logic has been reported in literature before [91,92]. In more detail, the distinction between product and process characteristics is based on Edquist et al. [93]: Product patents answer to the question *what*, while process patents answer to the question *how*. Keywords for incremental characteristics are based on the general innovation literature such as Helfat [30], Robertson et al. [94] and Klitkou et al. [24].

At this point, patents are assigned to a specific technological path within an operation unit and process (or product) as well as incremental (or radical) characteristics of these patents are identified. Thus, based upon the defined keywords reflecting process characteristics or an incremental degree of innovation, each patent is categorized as either product-radical, process-radical, product-incremental or process-incremental. We denote the total amount of product-radical patents by PR_t , process-radical patents by CR_t , product-incremental patents by PI_t , and process-incremental patents by CI_t within the patent portfolio. Based on the theoretical reasoning of path dependence [6,20,34,35,53,84], the path dependence indicator PD_t is introduced, which calculates the relationship between radical and incremental patent-types. More precisely, PD_t calculates, on an annual basis, the share between the sum of product-incremental and the process-incremental patents I_t with the sum of the product-radical and the process-radical patents R_t :

$$PD_t = \frac{PI_t + CI_t}{PR_t + CR_t} = \frac{I_t}{R_t}$$

Within step 3, the methodological approach follows the logic of the three-phase concept of Sydow et al. [6] based on the rationale that if low path dependence exists, patents reflecting product-radical and process-radical characteristics exceed those of product-incremental and process-incremental characteristics. Vis-à-vis, the opposite can be assumed for product, process, radical and incremental characteristics in the lock-in phase, when high path dependence exists [25,30,51]. Within the pre-formation phase, in product-oriented industries (e.g., automotive industry), mainly patents with product characteristics are applied for. This situation is different in process-oriented industries (e.g., oil and gas or chemical industry), where products usually require a specific process even in the early phase of development [87]. However, as a common characteristic in any industry, radical features outweigh incremental features in registered patents in the first phase due to unrestricted decision-making power that is not related to earlier decisions. Consequently, we assume that this leads to a PD_t value of ≤ 0.5 since the summarized shares in the denominator (product-radical and process-radical) exceed the summarized shares in the numerator

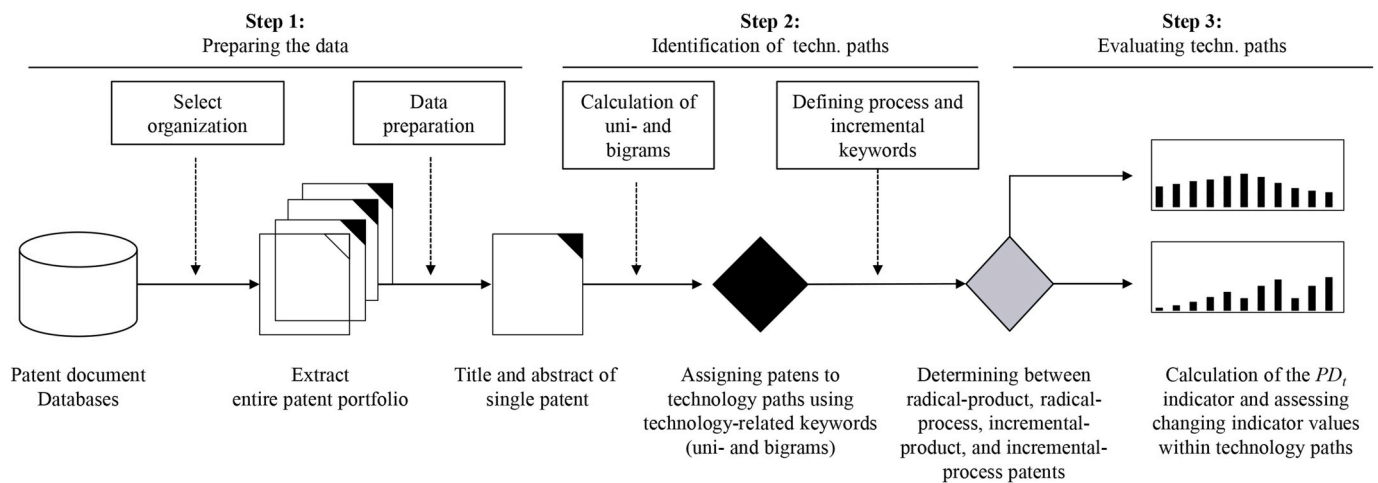


Fig. 2. Methodological approach distinguishes between three research steps.

Table 2

Keywords for identifying process and incremental patents.

Keywords for process patents	Keywords for incremental patents
method ^a for/of/to, process ^a for/of/to, workflow ^a for/of/to, procedure ^a for/of/to, proceeding ^a for/of/to, technique ^a for/of/to, approach ^a for/of/to, manufacturing ^a for/of/to, production ^a for/of/to	advanc ^a , advantag ^a , alternativ ^a , benefic ^a , better ^a , efficient ^a , enhanc ^a , expan ^a , extend ^a , improv ^a , increas ^a , increment ^a , modif ^a , reinforc ^a , superior ^a , support ^a , upgrad ^a

^a Note: The keywords consider both singular and plural.

(product-incremental and process-incremental) by a multiple (see Fig. 3). As the attractiveness of radical characteristics decreases in the formation phase, the total number of patents as well as patents with incremental characteristics depends on the chosen technology [95]. Thus, the PD_t value grows to a range between 0.5 and 1. A value within this range indicates that the summarized share in the numerator and denominator of PD_t is balanced. This shows that in this phase neither PR_t , nor CR_t , PI_t or CI_t dominate and vice versa. We assume that in the lock-in phase, incremental characteristics dominate the radical characteristics in technology developments [24,30]. Since the summarized shares of product-incremental and process-incremental patents exceed those of radical-product and radical-process patents, the value of PD_t continues to rise above 1. The fixed decision pattern holds a deterministic stimulus for all future events, leading to a decreasing number of total patents applied for. In this stage, alternative technological options are excluded due to the lock-in situation and innovation is mainly incremental and reinforces the extant path.

4. Results

We follow the proposed methodological approach and extract the entire patent portfolio of Royal Dutch Shell, BP, Chevron Corporation and ExxonMobil including their subsidiaries.⁴ More precisely, we select all INPADOC patent families with at least one granted patent published between 1970 and 2020. Each patent family is represented by one

⁴ We use the database Derwent Innovation to extract patents as it offers selecting assignees through a corporate tree (Assignee Code-DWPI), considering subsidiaries as well as mergers and acquisitions in assignee names. This applies only for granted US and EP patents as well as EP and WIPO patent applications.

granted patent, which serves as the unit of analysis throughout the study.⁵ We exclude those patents that do not contain a sufficient title, abstract and application date from the dataset.⁶ In total, we identify 10,611 patents for Royal Dutch Shell, 8849 patents for Chevron Corporation, 7818 patents for BP and 17,797 patents for ExxonMobil (see Fig. 4).

4.1. Assessing technological paths in Royal Dutch Shell

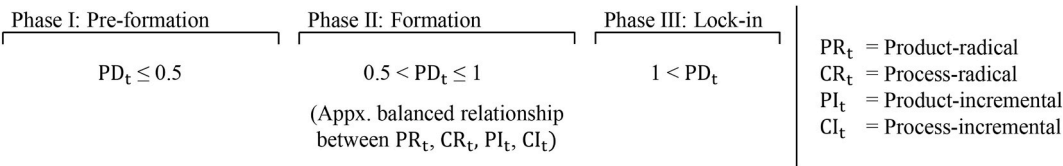
Exemplarily for Royal Dutch Shell, the n-gram based approach allows us to derive several hundred unigrams and bigrams, of which we create 105 unique and distinguishable technology-related keywords describing the three operation units: *Upstream* (23 unigrams, 7 bigrams), *Midstream* (12 unigrams, 5 bigrams) and *Downstream* (41 unigrams, 17 bigrams) (see Appendix Table A.3). The application of technology-related keywords assigns 7601 patents (72% of total patents) with an assignment level that exceeds the 60% threshold (see Fig. 5). A further 988 patents are assigned to more than one operation unit, and 2022 patents cannot be assigned based on the defined keyword scheme. These “unassignable” patents are part of the original company-specific dataset but do not contain sufficient keyword matches to allow a clear allocation to upstream, midstream, or downstream activities. We control the allocation process by means of an error analysis. Additionally, we observe that the application years of unassignable patents follow a similar pattern to those patents that can be assigned, suggesting that their exclusion does not systematically bias the results. Therefore, no further analysis is conducted on the unassignable patents.

To gain a deeper understanding of the multiple-technological paths that exist in each operation unit, we select the *Downstream* operation unit. This is the largest unit (by number of patents) which can be explained by a comparably higher degree of value added (i.e., specialty chemicals) as compared to bulk (commodity) products in the upstream operation unit. Within the annual report of Royal Dutch Shell [96], three major technological paths within the *Downstream* operation unit are described: *Fuel refinery*, *Gasification* and *Chemicals*.⁷ We used the most frequent uni- and bigrams as technology-related keywords to distinguish between different technological paths within the *Downstream* operation unit. We were able to assign 4310 patents (81%) to the three

⁵ Patents are summarized as INPADOC patent families. This means that within a family all patents claim the same priority while sharing the same invention. This approach minimizes double counting.

⁶ This affects approximately 5% of patents within each organization.

⁷ Fuel refinery also includes biofuels; upgrading bitumen are combined with chemicals.



PR_t = Product-radical

CR_t = Process-radical

PI_t = Product-incremental

CI_t = Process-incremental

Fig. 3. Three-phase model of path dependence reflected by patent behavior.

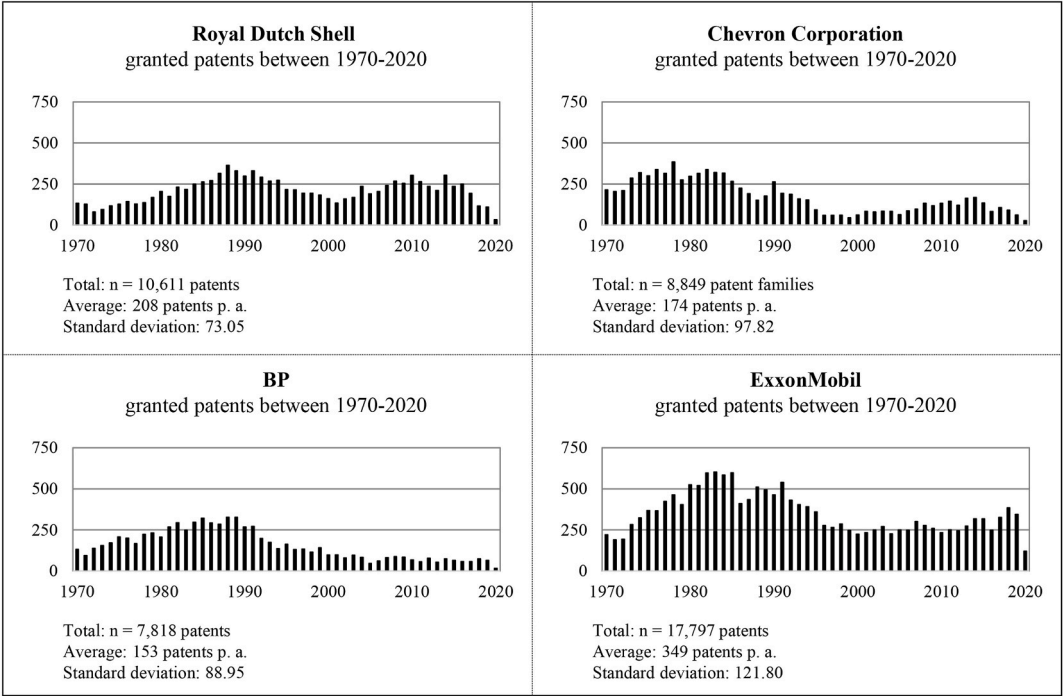


Fig. 4. Overview of granted patents for Royal Dutch Shell, Chevron Corporation, BP and ExxonMobil.

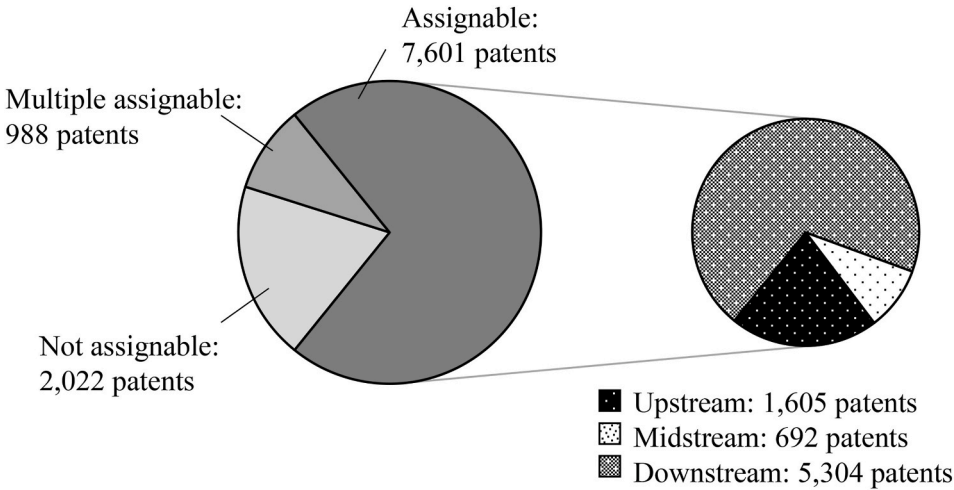


Fig. 5. Assignable patents for Royal Dutch Shell in accordance to *Upstream*, *Midstream* and *Downstream*.

technological paths and controlled the assignment process again by means of error analysis.⁸ In order to distinguish between product, process, radical and incremental characteristics, we apply the defined keywords to the titles and abstracts of the extracted patents, while

⁸ Out of the 4310 assignable patents, 3074 patents (58%) have a 100% accuracy assignment level, 133 patents (3%) have an 80-99% accuracy assignment level, 1103 patents (21%) have a 60-79% accuracy assignment level.

controlling for false positives and false negatives.⁹

4.2. Evaluating technological paths in Royal Dutch Shell

Overall, the methodological approach enables the empirical assessment of path dependence in technology developments of multi-technological organizations by using patents as a proxy. Fig. 6 shows the development of the different product, process, radical and incremental characteristics as well as the resulting PD_t values over the course of time. For the matter of clarity and visibility, we focus on the period from 1991 to 2020. In this period, 557 patents are assigned to the technological path of *Fuel Refinery*. Within this path, we observe a first peak of PD_t close to one in 1998 and two subsequent peaks close to 0.70 in 2009 and 2017. For the technological path of *Gasification*, we observe only one peak occurring in 2001, close to a value of 0.50. Most patents (1,262) were assigned to the technological path of *Chemicals*. Here, the PD_t value follows an overall (long term) trend with increasing PD_t values until 2011 as well as increasing fluctuation after 2008. We also observe a decreasing number of patents assigned to this specific technological path; compared to the numbers in 1991, the number of patents in 2003 decreases by 20%.

In general, we notice that most patents have process-radical characteristics. Therefore, Royal Dutch Shell seems to be a process-oriented company that constantly renews its processes within different technological paths. Based on the average PD_t values, the characteristics of overall filed patents and the fact that the radical characteristics (PR_t or CR_t) are dominating incremental characteristics (PI_t and CI_t), the *Downstream* activities in the technological path *Fuel refinery*, *Gasification* and *Chemicals* of Royal Dutch Shell are assigned to either phase I, the pre-formation phase, or phase II, the formation phase (compare Figs. 1 and 3). This leads to the assumption that Shell realizes scale, scope and learning effects within most technological paths in the *Downstream* operation unit. The moderate PD_t value especially in the last years implies that the organization is still able to adopt novel opportunities for each technological path.

To support monitoring and further evaluating the developments of the technological paths, a distinction between PI_t , CI_t , PR_t and CR_t is advisable. As illustrated in Fig. 7, Royal Dutch Shell's portfolio underlines the company's character of belonging to the process industry. Further, we see that most patents include non-incremental characteristics, and thus, we assume that the company is still able to renew its portfolio within the existing technology paths. Especially, within the technology path *Gasification* we see a development to more process-radical patents over the course of the last centuries.

4.3. Discussion and comparison of the proposed methodological approach

To investigate whether our proposed methodological approach applies not only to Royal Dutch Shell but also to other multi-technological organizations in the oil and gas industry, we apply our approach to Chevron Corporation, BP and Exxon Mobil. We observe that, overall, all three competitors have a similar patent portfolio structure with similar assigning rates compared to Royal Dutch Shell (see Tables 3 and 4).

While looking at the activities of the largest operation unit (by number of patents), that is *Downstream*, for the four selected companies all three technological paths can be identified with *Chemicals* being the most patented path in all patent portfolios. Following the argument that not only internal but also external mechanisms, such as coordination and complementary effects [20,97], cause path dependence, we notice that these mechanisms seem to have an effect on at least two of the selected organizations in the industry (i.e., we identify a similar pattern

for Chevron Corporation and ExxonMobil, exemplarily illustrated for the path *Fuel refinery* in Fig. 8). This is already stated in literature for the oil and gas industry, described as socio-technical structures and industrial locked-in patterns [see e.g., 70–73]. However, we are aware that the sample size (i.e., two companies within the industry) is too small to provide sufficient statistical evidence and outline this as a direction for future research. Still, our results indicate that most technological paths within the *Downstream* operation units of all four organizations belong to either phase I, the pre-formation phase, or phase II, the formation phase.

Next, we focus on the time period 2003–2020, as 2003 indicates the publishing year of the European biofuels directive that targets at implementing a minimum proportion of biofuels and other renewable fuels on the market [98]. We are interested whether adopting new technologies (that might lead to new technological paths) is more difficult for organizations with higher PD_t values than for those organizations with lower PD_t values. We draw upon organizations' opportunities on adopting bio-based energy (or fuels) and chemical activities and, therefore, identify biomass related patents¹⁰ within the technological paths of *Fuel refinery* and *Chemicals* for all companies.

In the period 2003–2020, we identified 40 biomass patents within *Fuel refining* and 14 biomass patents within *Chemicals* for Royal Dutch Shell, while we identified seven biomass patents within *Fuel refining* and nine biomass patents within *Chemicals* for Chevron Corporation. Further, we found 13 biomass patents within *Fuel refining* and two biomass patents within *Chemicals* for BP, and seven biomass patents within *Fuel refining* and four biomass patents within *Chemicals* for ExxonMobil.

Exemplarily for the technological path *Fuel refinery*, Fig. 9 shows the development of the patent portfolios from 2003 to 2020 as well as the influence of biomass related patents. It seems that the biomass patents go along with a shift towards a more radical patent portfolio for each company within the *Fuel refinery* technology path. Interestingly, the *Fuel refinery* technology path of BP seems to be affected most by biomass related patents, also due to the highest share of biomass patents in this path for BP.

To further compare our results and consider strategic alliances, partnerships or joint ventures that are not reflected in an organizations' patent portfolio, we analyzed the at the time of analysis most recent annual reports (i.e., 2019) and corporate investments of the four companies based on Pitchbook data (see Table 5). Pitchbook provides company, investment, and funding data. Based on the annual reports, we investigate the company's vision as well as strategy regarding sustainability. Thereby, especially Royal Dutch Shell and BP state that they are willing to invest in a positive contribution to the energy transition and to reduce emissions. Further, Royal Dutch Shell integrates sustainability actions within its operations, customers, and communities. As stated in their reports, BP also focuses on new technologies such as biofuels, wind energy or renewable products to provide a new energy mix with reduced carbon. Contrary, ExxonMobil states that they stick to ethical standards while meeting the energy demand. In the same way, Chevron Corporation aims to reach operational excellence without specifically mentioning sustainability in their corporate strategy. But Chevron Corporation is integrated in projects involving renewable resources as well as reduction of fresh water. The analysis of Pitchbook data reveals several acquisitions and investments in general and some investments specifically in CleanTech and ClimateTech. Especially, Royal Dutch Shell and BP show a high number of investments within renewable energy while for ExxonMobil and Chevron Corporation less than five investments within these fields are listed within the data base. To a certain extent, these numbers are reflected in the number of biomass related patents in the technological path of fuel refinery as well as in the entire

⁹ We manually tested a random sample of 371 patents (desired confidence level 95%). The test indicates a false positive indication of 7.7% and a false negative indication of 9.2% for process patents.

¹⁰ We use the keywords biomass*, cellulose* and lignocellulos* to identify biomass related patents.

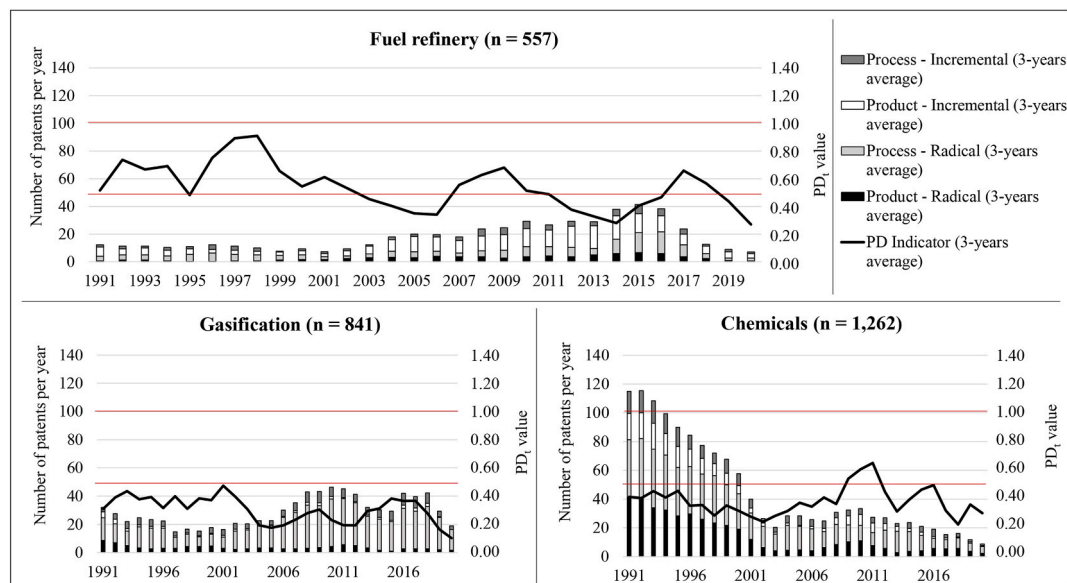


Fig. 6. Overview of development of Product-Radical (PR), Process-Radical (CR), Product-Incremental (PI) and Process Incremental (CI) patents in respect to technological paths of Downstream activities for Royal Dutch Shell. Note: patents are sorted by application year.

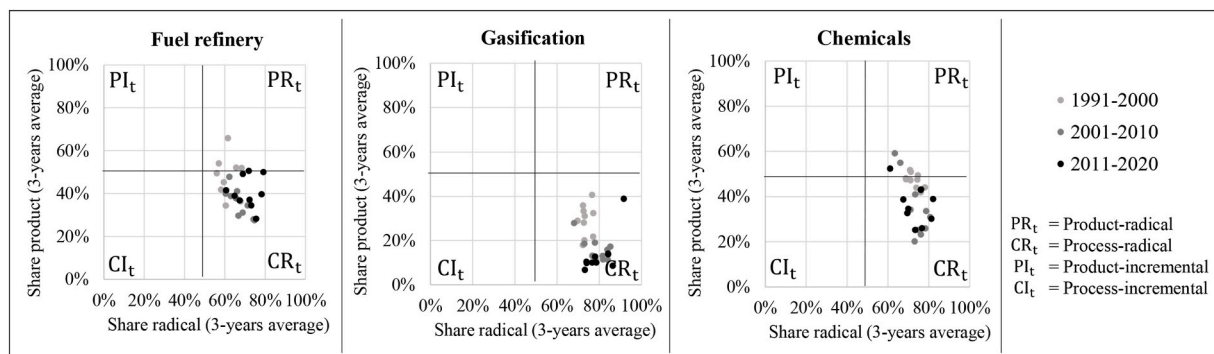


Fig. 7. Overview of development of portfolio in respect to technological paths of Downstream activities for Royal Dutch Shell.

Table 3

Organizations portfolio of patents: Upstream, Midstream and Downstream.

Organizations	Patents between 1970 and 2020				
	Total patents	Assigning rate	Upstream	Midstream	Downstream
Royal Dutch Shell	10,611	72%	1605 (15%)	692 (7%)	5304 (50%)
Chevron Corporation	8849	68%	1501 (17%)	556 (6%)	3978 (45%)
BP	7818	71%	1556 (20%)	465 (6%)	3523 (45%)
ExxonMobil	17,797	72%	2348 (13%)	865 (5%)	9579 (54%)

Note: The assigning rates exclude 1646 (19%) patents of Chevron Corporation, 1534 (20%) patents of BP and 3174 (18%) patents of ExxonMobil that were not assignable (accuracy level below the 60% threshold). Additionally, the assigning rates exclude 1168 (13%) patents of Chevron Corporation, 740 (9%) patents of BP and 1831 (10%) patents of ExxonMobil that are assigned to multiple paths.

patent portfolio of each company.

To conclude, based on the PD_t values we see that Royal Dutch Shell seems to be the least path dependent company which is also reflected within several CleanTech and Cimatech investment activities of this company as well as potential path breaking behavior by applying for biomass related patents. Therefore, we assume that the presented PD_t indicator seems to give a realistic picture of path dependence in multi-technological organizations.

5. Conclusion

In this study, we develop and apply a novel measure to empirically assess path dependence in distinctive technology developments of multi-technological organizations. We provide empirical evidence for the emergence of technological path dependence by characterizing patents of oil- and gas-producing organizations. To do so, we distinguish between product-radical, process-radical, product-incremental and

Table 4
Organizations portfolio of patents: Fuel refinery, Gasification and Chemicals.

Organizations	Patents between 1970 and 2020				
	Total patents	Assigning rate	Fuel refinery	Gasification	Chemicals
Royal Dutch Shell	5304	81%	748 (14%)	1286 (24%)	2276 (43%)
Chevron Corporation	3978	76%	949 (24%)	761 (19%)	1331 (33%)
BP	3523	81%	404 (11%)	692 (20%)	1769 (50%)
ExxonMobil	9579	77%	1451 (15%)	1216 (13%)	4739 (49%)

Note: The assigning rates exclude patents that were assignable but with an accuracy assignment level below the 60% threshold.

process-incremental patents and present a novel indicator (PD_t). The application of the methodological approach to the patent portfolio of Royal Dutch Shell, Chevron Corporation, BP and Exxon Mobil delivers insights on how path dependence emerged within the different technological paths.

Next to the methodological advancements in empirically assessing path dependence in multi-technological organizations, our approach offers several theoretical contributions. First, a comparison of PD_t empirically supports the argument that the mechanisms of path dependence are dependent on contextual factors such as the surrounding industry [18,25,34,35,99]. However, examination of technologies related to biomass as well as corporate investments in CleanTech and Cimatech shows that the adoption of emerging technologies potentially leading to new technological paths is higher in those organizations with lower PD_t values giving insights into potential path breaking initiatives. Although this approach should not be interpreted as a robust validation in a strict methodological sense, we observe and argue that organizations with a comparably lower PD_t do not show the same patterns as their peers. Thus, they seem not to be affected by potential path dependence mechanisms that are part of the surrounding industry. This finding is particular important in the current political debate for building a strong bio-based economy with the ambition to foster sustainability transitions [100,101]. Second, the approach extends qualitative research designs of Ibrus [23], Klitkou et al. [24] and Stache and

Sydow [62] through offering a basis for verifying individuals' perception of path dependence. This supports to better understand whether path dependence in distinctive technology developments is reflected by managerial imprint or whether it is rather a phenomenon of an entire industry, as for example discussed in Bergek and Onufrey [20] and Alkaersig et al. [25]. Third, the presented approach underscores previous findings demonstrating that in organizations, path dependence refers to specific technological paths and that several different paths co-exist in multi-technological organization [20,35].

From a practitioner's perspective, our approach enables organizations to monitor their technology base and to raise awareness for adopting path-breaking technologies. Specifically, our approach is of value for monitoring as it, first of all, allows for identifying different technological paths and, thus, for comparing potential degrees of path dependence across the extant technology portfolio (of the company in question but also as a means of monitoring its competitors). This provides a basis for more proactive strategic adjustments, and, thus, aiding the orchestration of technology developments across the technology portfolio. By focusing on the case at hand, a direct comparison of PD_t for all four oil and gas organizations, the technological trajectories in most of the *Downstream* operation units are consistent with the characteristics of the pre-formation or formation phase. Companies in this stage realize scale, scope, and learning effects within the different technological paths. However, especially those companies with high PD_t values need to closely monitor the latest developments to react if the score grows until signaling a lock-in phase.

Nevertheless, future research is needed to overcome some shortcomings of the proposed methodological approach. One limitation of the current approach relates to the scope of the investigation. As it is currently designed, the indicator measures the extent of technological concentration and branching within the set of technological paths represented in a firm's patent portfolio. It therefore identifies patterns of reinforcement and lock-in within observable technological domains. However, it does not directly measure dependence on technologies absent from the portfolio. Consequently, its ability to detect vulnerability to fundamentally external or disruptive technological trajectories is limited. Future research could therefore examine firm-level patent portfolios within a broader technological landscape. While the current framework captures structural reinforcement patterns within observed technological activities, it does not assess path dependence relative to

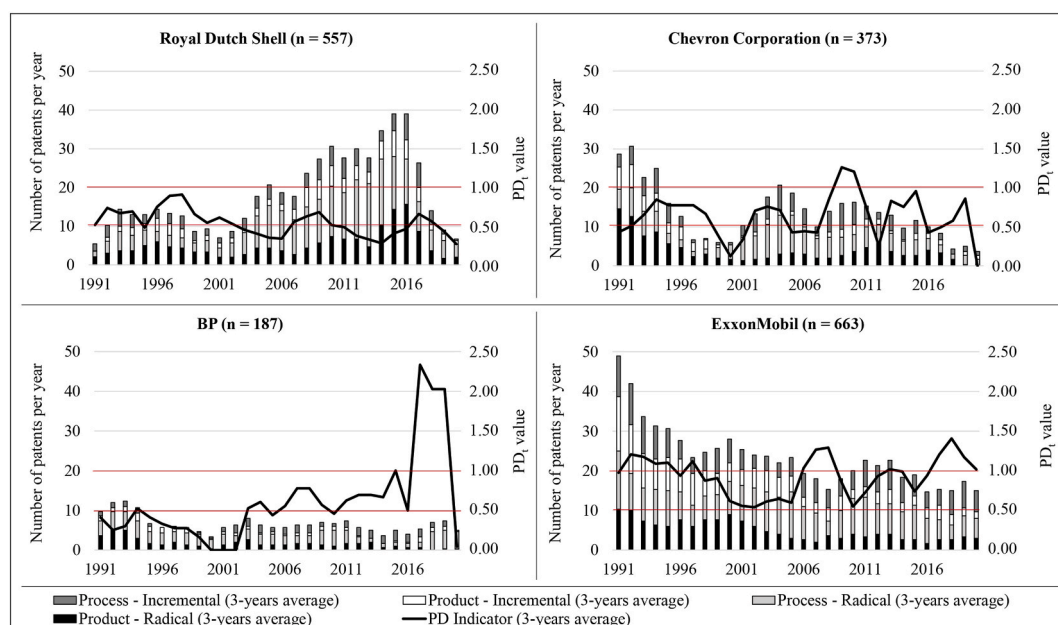


Fig. 8. Overview on development of PD_t value in the path *Fuel refinery* within the four different organizations.

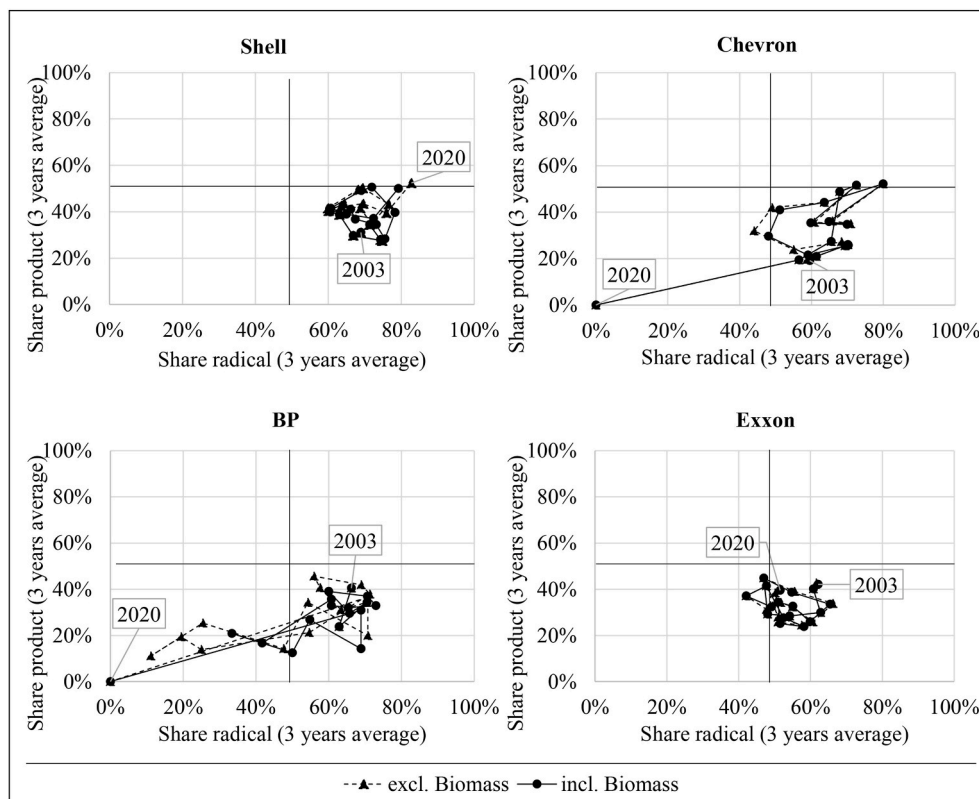


Fig. 9. Overview of portfolio development in respect to biomass patents within technology path *Fuel refinery*.

Table 5

Comparison analysis of average PD_t for Royal Dutch Shell, Chevron Corporation, BP and ExxonMobil for the technology paths of Fuel Refinery between 2003 and 2020, biomass related patents as well as investments in CleanTech and ClimateTech. P_t shows the number of all patents PB_t the number of biomass related patents in fuel refinery.

	Fuel refinery			
	Royal Dutch Shell	Chevron Corporation	BP	Exxon Mobil
P_t (entire portfolio)	419 (3815)	209 (1903)	100 (1211)	330 (4904)
PB_t (biomass patents in fuel refinery biomass patents in entire portfolio)	10% (40 163)	3% (7 24)	13% (13 32)	2% (7 25)
PD_t	0.45	0.71	0.87	0.97
Investments (in CleanTech and ClimateTech)	106 (30)	38 (2)	112 (17)	51 (4)

emerging external trajectories or potentially disruptive paradigms. Extending the approach to incorporate developments across industry sectors would enable both internal technological path dependence and firms' relative positioning in relation to emerging technological fields to be examined. In this context, a layered or hierarchical perspective could be useful for distinguishing between (1) intra-field technological path dependence and (2) cross-field or paradigm-level path dependence. Furthermore, future studies should carefully interpret the small patent portfolios of "alien" technologies, as these patents may be used by firms to prevent other firms from developing technologies that could potentially disrupt their current business. Such patterns would require careful interpretation in future studies on path dependence and could provide further evidence of the path dependence behavior of the firms under investigation.

Additional limitations are mainly related to the methodological

approach used in this paper. Clearly, the selected thresholds of the three-phase model can only be vaguely defined at present, which could have the side-effect of limiting our ability to characterize transitions between phases. However, organizations themselves can also set their individual threshold value. Further, patent data go along with some general limitations. For instance, emerging technologies may be misinterpreted if assessed on single data sources [102,103]. Additionally, patent data only reflect upon each organization's own R&D activities, while other activities, such as collaborations with start-ups, as well as their performance are not represented. Notably, the process of assigning patents to technology paths can be further improved by extending the number of technology-related keywords that are used. However, the assigned threshold level of 60% may need to be adapted accordingly. This is also important vis-à-vis the ability to make distinctions between certain technology keywords so that the rate of non-assignable patents might decrease even further. This would also allow for a more fine-grained differentiation between the relevant sub-technological pathways for *Fuel refinery*, *Gasification*, and *Chemicals*. While ongoing advances in semantic text analysis may enable the development of more refined indicators, our approach is limited by relying on patent titles and abstracts. Future research could incorporate full patent descriptions and claims to capture the technological substance and legal scope of inventions more precisely, particularly when distinguishing between incremental and radical innovations. In addition, parameters reflecting industry-specific characteristics, such as the relative importance of process versus product innovations, could further refine the indicator. Furthermore, as we are aware that the sample size is too low to determine whether a relationship between the adoption rate of biomass-related technologies or the amount of mergers and acquisitions in clean technologies and the value of the low PD_t indicator exists, future research needs to extend the approach by, for example, applying it to additional organizations within or outside the oil and gas industry. Such comparisons with other data sources should also not be interpreted as a robust validation in a strict methodological sense. A validation would

require, for example, a comparison with an independent benchmark. Moreover, the operationalization of patents as a proxy for technology developments is itself subject to some specific limitations. It should be considered, for instance, that the existence of patenting standards acts, in some sense, as a natural barrier to innovation, as every patent application must demonstrate that the invention features a certain degree of novelty, non-obviousness, and utility. Thus, not all inventions can be patented, which might be an obstacle for incremental innovation. Therefore, we are aware that patent data does not necessarily reflect the pace and scope of technology developments, which also applies to the present approach. However, future research could potentially make use of the foregoing research design in the context of studies at the level of decision makers in companies [55,104,105] or their financial performance [106,107] in order to gain a comprehensive understanding how path dependence emerges in and affects multi-technological organization.

Appendix

Table A.1
Derived keywords for the operation units Upstream, Midstream and Downstream

Upstream	Midstream	Downstream
float	pipe	Fischer Tropsch
marine	distribution	mineral oil
offshore	sulfur	heating oil
seismic	liquefying gas	hydrocarbon cooling
wellhead	conductor	hydrocarbon cracking
Sea	transport	gasoil
subsea	storage	diesel
oil well	tube	petroleum
onshore	separator	gasoline
deep water	compressor	kerosene
ocean	liquefaction	furnace
offshore	terminal	hydrotreating
platform	oil processing	fuel
tension leg	gas processing	hydrocarbon fuel
soil	reservoir	hydrocarbon oil
terra	integrated gas	catalytic cracking
CO2	liquefied natural gas	octane
crude		naphtha
drill		bitum
bore		solid fuel
oil rig		oxygenate
extract bitumen		hydrocarbon gas
oil shale		gasification
synthetic oil		H2
extract oil		propane
sand		ethane
ship		butane
surface		methane
casing		carbon monoxide
cementing		carbon dioxide
		syngas
		synthesis gas
		hydrogen fuel
		hydrogen oil
		hydrogen gas
		natural gas
		ethylen
		propylene
		butadiene
		aromatic
		olefin
		hydroformylation
		styrene
		phenol
		asphalt
		ammonia
		elastom

(continued on next page)

CRediT authorship contribution statement

Stefanie Bröring: Writing – review & editing, Funding acquisition, Conceptualization. **Anna Waßenhoven:** Writing – review & editing, Visualization, Validation, Formal analysis, Data curation. **Silvan Berg:** Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Michael Rennings:** Writing – review & editing, Visualization, Validation, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table A.1 (continued)

Upstream	Midstream	Downstream
		carboxylic methanol polystyrene lubrica epoxy polymer aralkyl polyketone thermoplastic monomer pesticide

Table A.2

Derived keywords for the technology paths Fuel refinery, Gasification and Chemicals

Fuel refinery	Gasification	Chemicals
Fischer Tropsch	hydrocarbon gas	aromatic
mineral oil	gasification	olefin
heating oil	H ₂	hydroformylation
hydrocarbon cooling	propane	styrene
hydrocarbon cracking	ethane	phenol
gasoil	butane	asphalt
diesel	methane	ammonia
petroleum	carbon monoxide	elastom
gasoline	carbon dioxide	carboxylic
kerosene	syngas	methanol
furnace	synthesis gas	polystyrene
hydrotreating	hydrogen fuel	lubrica
fuel	hydrogen oil	epoxy
hydrocarbon fuel	hydrogen gas	polymer
hydrocarbon oil	natural gas	aralkyl
catalytic cracking	ethylen	polyketone
octane	propylene	thermoplastic
naphtha	butadiene	monomer
bitum		pesticide
solid fuel		
oxygenate		

Table A.3

Examples how the process specific keywords help to identify process characteristics in patents

Keyword	Patent no.	Sample text excerpt from the patent title or abstract
method for	EP2794810B1	The present invention provides a method for enhancing the recovery of oil from a formation [...]
methods for	US9580602B2	Methods for digesting cellulosic biomass solids may comprise: providing cellulosic biomass solids in the presence of a digestion solvent [...]
process for	EP2981515B1	The invention provides a process for the preparation of ethylene glycol and 1, 2-propylene glycol [...]
method of	EP2451890B1	A method of treating a formation containing crude oil with specific solubility groups and chemical families is described [...]
process of	US10017432B2	The invention relates to a process of the oxidative dehydrogenation of an alkane containing 2 to 6 carbon atoms [...]
methods to	US7335802B2	Systems and methods to produced branched aliphatic alcohols are described [...]
process to	US9487400B2	A process to prepare a synthesis gas mixture comprising hydrogen and carbon monoxide from a liquid hydrocarbon feedstock [...]

Table A.4

Examples how the incremental specific keywords help to identify incremental characteristics in patents

Keyword	Patent no.	Sample text excerpt from the patent title or abstract
advanc*	EP302560A3	A process is disclosed for preparing an advanced epoxy resin by reacting a relatively low molecular weight polyepoxide with a phenolic compound [...]
advantag*	EP984826B2	Processes using autothermic modules in accordance with this invention are advantageously used for production of synthesis gas comprising carbon monoxide [...]
alternative*	US6232261B1	A catalyst system suitable for the polymerization of olefins [...]. The activators provide an alternative to the traditional use of aluminoxanes alone.
benefic*	US5324702A	[...] Carbon dioxide used as a carrier gas with a methylaromatic feed to the oxidation catalyst is shown to have a beneficial effect on yield and selectivity.
better*	US4081491A	The preparation of solid saturated epoxy resin compositions [...]. The preparation of solid saturated epoxy resin compositions [...]
efficient*	EP928326B1	The fuel composition used in the inventive method permits a safe and highly efficient operation of a dry low NO _x combustion system [...]
enhanc*	EP2794810B1	The present invention provides a method for enhancing the recovery of oil from a formation
expan*	EP969037B1	The present invention relates to a homogeneous composition of an expandable styrene polymer in the form of beads comprising [...]
extend*	US5241114A	A method and a system for extending useful life of a basic catalyst are disclosed [...]

(continued on next page)

Table A.4 (continued)

Keyword	Patent no.	Sample text excerpt from the patent title or abstract
improv*	US9528052B2	A process that provides for the improvement of the properties of a distillate feedstock [...]
increas*	US8926716B2	In a fuel composition containing a static dissipator additive, a Fischer-Tropsch derived fuel component is blended for the purpose of increasing the electrical conductivity of the composition [...]
increment*	US6166134A	A known flow rate of the gas is added to the combined liquid [...] and the incremental change in concentration of the gas in the recirculating vapor flowstream [...]
modif*	EP1414869B1	The process and the device are particularly suitable for continuously manufacturing a styrene polymer, a styrene copolymer having a high impact strength modified by grafting onto a rubber [...]
reinforc*	US4465797A	A reinforced polymer composition comprising an epoxy resin matrix having intimately distributed [...]
superior*	US9580666B2	The lubricating oil composition of the present invention has a superior friction reducing effect and which has a superior fuel economy effect.
support*	US7473814B2	The metal catalyst may be chosen from metal catalysts supported on and preferably grafted to a solid support. One of the main advantages of the present invention is to produce ethane with a very high selectivity.
upgrad*	US7261808B2	A process to further upgrade a pre-processed used lubricating oil [...]

Data availability

Data will be made available on request.

References

- [1] N. Goyal, M. Howlett, Who learns what in sustainability transitions? *Environ. Innov. Soc. Transit.* 34 (2020) 311–321, <https://doi.org/10.1016/j.eist.2019.09.002>.
- [2] F.W. Geels, The multi-level perspective on sustainability transitions: responses to seven criticisms, *Environ. Innov. Soc. Transit.* 1 (2011) 24–40, <https://doi.org/10.1016/j.eist.2011.02.002>.
- [3] F.W. Geels, Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study, *Res. Pol.* 31 (2002) 1257–1274, [https://doi.org/10.1016/S0048-7333\(02\)00062-8](https://doi.org/10.1016/S0048-7333(02)00062-8).
- [4] T. Magnusson, V. Werner, Conceptualisations of incumbent firms in sustainability transitions: insights from organisation theory and a systematic literature review, *Bus. Strat. Environ.* 32 (2023) 903–919, <https://doi.org/10.1002/bse.3081>.
- [5] R. Bohnsack, H. Kurtz, A. Hanelt, Re-examining path dependence in the digital age: the evolution of connected car business models, *Res. Pol.* 50 (2021) 104328, <https://doi.org/10.1016/j.respol.2021.104328>.
- [6] J. Sydow, G. Schreyögg, J. Koch, Organizational path dependence: opening the black box, *Acad. Manag. Rev.* 34 (2009) 689–709, <https://doi.org/10.5465/amr.34.4.zok689>.
- [7] M. Lawrence, Taking stock of the ability to change: the effect of prior experience, *Organ. Sci.* 29 (2018) 489–506, <https://doi.org/10.1287/orsc.2017.1181>.
- [8] W.B. Arthur, Competing technologies, increasing returns, and Lock-In by historical events, *Econ. J.* 99 (1989) 116, <https://doi.org/10.2307/2234208>.
- [9] K.E. Boulding, What is evolutionary economics? *J. Evol. Econ.* 1 (1991) 9–17, <https://doi.org/10.1007/BF01202334>.
- [10] M. Kenney, E. von Burg, Technology, entrepreneurship and path dependence: industrial clustering in silicon valley and route 128, *Ind. Corp. Change* 8 (1999) 67–103, <https://doi.org/10.1093/icc/8.1.67>.
- [11] R.W. Rycroft, D.E. Kash, Path dependence in the innovation of complex technologies, *Technol. Anal. Strat. Manag.* 14 (2002) 21–35, <https://doi.org/10.1080/09537320220125865>.
- [12] P. Aghion, Path dependence, innovation and the economics of climate change, in: R. Fouquet (Ed.), *Handbook on Green Growth*, Edward Elgar Publishing, 2019, pp. 67–83.
- [13] A. Schmitt, S. Raisch, H.W. Volberda, Strategic renewal: past research, theoretical tensions and future challenges, *Int. J. Manag. Rev.* 20 (2018) 81–98, <https://doi.org/10.1111/ijmr.12117>.
- [14] G. Schreyögg, J. Sydow, P. Holtmann, How history matters in organisations: the case of path dependence, *Manag. Organ. Hist.* 6 (2011) 81–100, <https://doi.org/10.1177/1744935910387030>.
- [15] R. Agarwal, C.E. Helfat, Strategic renewal of organizations, *Organ. Sci.* 20 (2009) 281–293.
- [16] M. Gruber, I.C. MacMillan, J.D. Thompson, Escaping the prior knowledge corridor: what shapes the number and variety of market opportunities identified before market entry of technology start-ups? *Organ. Sci.* (2013) 280–300.
- [17] R. Simons, *Levers of Control: How Managers Use Innovative Control Systems to Drive Strategic Renewal*, Harvard Business School Press, Boston, 1995.
- [18] K. Onufrey, A. Bergek, Transformation in a mature industry: the role of business and innovation strategies, *Technovation* 105 (2021) 102190, <https://doi.org/10.1016/j.technovation.2020.102190>.
- [19] O. Granstrand, S. Sjölander, Managing innovation in multi-technology corporations, *Res. Pol.* 19 (1990) 35–60, [https://doi.org/10.1016/0048-7333\(90\)90033-3](https://doi.org/10.1016/0048-7333(90)90033-3).
- [20] A. Bergek, K. Onufrey, Is one path enough? Multiple paths and path interaction as an extension of path dependency theory, *Ind. Corp. Change* 23 (2014) 1261–1297, <https://doi.org/10.1093/icc/dtt040>.
- [21] J. Fagerberg, Mobilizing innovation for sustainability transitions: a comment on transformative innovation policy, *Res. Pol.* 47 (2018) 1568–1576, <https://doi.org/10.1016/j.respol.2018.08.012>.
- [22] P.C. Bhatt, K.-K. Lai, V.A. Drave, T.-C. Lu, V. Kumar, Patent analysis based technology innovation assessment with the lens of disruptive innovation theory: a case of blockchain technological trajectories, *Technol. Forecast. Soc. Change* 196 (2023) 122864, <https://doi.org/10.1016/j.techfore.2023.122864>.
- [23] I. Ibrus, Path dependencies in media design: evolutionary dynamics of early mobile web and its textual forms, *Soc. Semiot.* 24 (2014) 191–208, <https://doi.org/10.1080/10350330.2013.859479>.
- [24] A. Klitkou, S. Bolwig, T. Hansen, N. Wessberg, The role of lock-in mechanisms in transition processes: the case of energy for road transport, *Environ. Innov. Soc. Transit.* 16 (2015) 22–37, <https://doi.org/10.1016/j.eist.2015.07.005>.
- [25] L. Alkaersig, K. Beukel, G. Laut, The effect of path-dependence and uncertainty on the value of mature technologies, *Int. J. Innovat. Manag.* 22 (2018) 1–32, <https://doi.org/10.1142/S1363919618500056>.
- [26] A. Keller, S. Konlechner, W.H. Güttel, G. Reischauer, Overcoming path-dependent dynamic capabilities, *Strateg. Organ.* (2022) 147612702211258, <https://doi.org/10.1177/14761270221125807>.
- [27] D. Geiger, E. Antonacopoulou, Narratives and organizational dynamics, *J. Appl. Behav. Sci.* 45 (2009) 411–436, <https://doi.org/10.1177/0021886309336402>.
- [28] C. Schubert, J. Sydow, A. Windeler, The means of managing momentum: bridging technological paths and organisational fields, *Res. Pol.* 42 (2013) 1389–1405, <https://doi.org/10.1016/j.respol.2013.04.004>.
- [29] Q. Plantec, P. Le Masson, B. Weil, Impact of knowledge search practices on the originality of inventions: a study in the oil & gas industry through dynamic patent analysis, *Technol. Forecast. Soc. Change* 168 (2021) 120782, <https://doi.org/10.1016/j.techfore.2021.120782>.
- [30] C.E. Helfat, Evolutionary trajectories in petroleum firm R&D, *Manag. Sci.* 40 (1994) 1720–1747, <https://doi.org/10.1287/mnsc.40.12.1720>.
- [31] T.H. Moran, Managing an oligopoly of would-be sovereigns: the dynamics of joint control and self-control in the international oil industry past, present, and future, *Int. Organ.* 41 (1987) 575–607, <https://doi.org/10.1017/S0020818300027612>.
- [32] P.I. Davidsen, J.D. Sterman, G.P. Richardson, A petroleum life cycle model for the United States with endogenous technology, exploration, recovery, and demand, *Syst. Dyn. Rev.* 6 (1990) 66–93, <https://doi.org/10.1002/sdr.4260060105>.
- [33] M.C. Simoens, S. Leipold, L. Fuenfschilling, Locked in unsustainability: understanding lock-ins and their interactions using the case of food packaging, *Environ. Innov. Soc. Transit.* 45 (2022) 14–29, <https://doi.org/10.1016/j.eist.2022.08.005>.
- [34] L. Dobusch, E. Schussler, Theorizing path dependence: a review of positive feedback mechanisms in technology markets, regional clusters, and organizations, *Ind. Corp. Change* 22 (2013) 617–647, <https://doi.org/10.1093/icc/dts029>.
- [35] K. Onufrey, A. Bergek, Self-reinforcing mechanisms in a multi-technology industry: understanding sustained technological variety in a context of path dependency, *Ind. Innovat.* 22 (2015) 523–551, <https://doi.org/10.1080/13662716.2015.1100532>.
- [36] J.P. Eggers, A. Kaul, Motivation and ability? A behavioral perspective on the pursuit of radical invention in multi-technology incumbents, *Acad. Manag. J.* 61 (2018) 67–93, <https://doi.org/10.5465/amj.2015.1123>.
- [37] L.M. Klüppel, L. Pierce, J.A. Snyder, Perspective—the deep historical roots of organization and strategy: traumatic shocks, culture, and institutions, *Organ. Sci.* 29 (2018) 702–721, <https://doi.org/10.1287/orsc.2017.1173>.
- [38] K. Zhu, K.L. Kraemer, V. Gurbaxani, S.X. Xu, Migration to open-standard interorganizational systems: network effects, switching costs, and path dependency, *MIS Q.* (2006) 515–539, <https://doi.org/10.2307/25148771>.
- [39] S. Karim, A. Kaul, Structural recombination and innovation: unlocking intraorganizational knowledge synergy through structural change, *Organ. Sci.* 26 (2015) 439–455, <https://doi.org/10.1287/orsc.2014.0952>.
- [40] G. Schreyögg, J. Sydow, CROSSROADS—Organizing for fluidity? Dilemmas of new organizational forms, *Organ. Sci.* 21 (2010) 1251–1262, <https://doi.org/10.1287/orsc.1100.0561>.

- [41] I. Dierickx, K. Cool, Asset stock accumulation and sustainability of competitive advantage, *Manag. Sci.* 35 (1989) 1504–1511, <https://doi.org/10.1287/mnsc.35.12.1504>.
- [42] F.W. Geels, From sectoral systems of innovation to socio-technical systems, *Res. Pol.* 33 (2004) 897–920, <https://doi.org/10.1016/j.respol.2004.01.015>.
- [43] W. Kremsler, G. Schreyögg, The dynamics of interrelated routines: introducing the cluster level, *Organ. Sci.* 27 (2016) 698–721, <https://doi.org/10.1287/orsc.2015.1042>.
- [44] J. Hepp, Being small at the right moment: path dependence after a shift in the technological regime, *Ind. Corp. Change* 31 (2022) 464–499, <https://doi.org/10.1093/icc/dtac001>.
- [45] C. Gärtner, O. Schön, Modularizing business models: between strategic flexibility and path dependence, *J. Strateg. Manag.* 9 (2016) 39–57, <https://doi.org/10.1108/JMSA-12-2014-0096>.
- [46] S. Heil, T. Bornemann, Creating shareholder value via collaborative innovation: the role of industry and resource alignment in knowledge exploration, *R D Manag.* 48 (2018) 394–409, <https://doi.org/10.1111/radm.12258>.
- [47] J.R. Lincoln, D. Guillot, M. Sargent, Business groups, networks, and embeddedness: innovation and implementation alliances in Japanese electronics, 1985–1998, *Ind. Corp. Change* 26 (2017) 357–378, <https://doi.org/10.1093/icc/dtw037>.
- [48] S. Kaplan, R. Henderson, Inertia and incentives: bridging organizational economics and organizational theory, *Organ. Sci.* 16 (2005) 509–521, <https://doi.org/10.1287/orsc.1050.0154>.
- [49] M. Tripsas, G. Gavetti, Capabilities, cognition, and inertia: evidence from digital imaging, *Strateg. Manag. J.* 21 (2000) 1147–1161.
- [50] P. Pierson, Increasing returns, path dependence, and the study of politics, *Am. Polit. Sci. Rev.* 94 (2000) 251–267, <https://doi.org/10.2307/2586011>.
- [51] S. Thrane, S. Blaabjerg, R.H. Møller, Innovative path dependence: making sense of product and service innovation in path dependent innovation processes, *Res. Pol.* 39 (2010) 932–944, <https://doi.org/10.1016/j.respol.2010.04.003>.
- [52] J. Woiceshyn, Integrative capability and technology adoption: evidence from oil firms, *Ind. Corp. Change* 14 (2005) 307–342, <https://doi.org/10.1093/icc/dth053>.
- [53] J. Sydow, A. Windeler, C. Schubert, G. Möllering, Organizing R&D consortia for path creation and extension: the case of semiconductor manufacturing technologies, *Organ. Stud.* 33 (2012) 907–936, <https://doi.org/10.1177/0170840612448029>.
- [54] J.-P. Vergne, R. Durand, The path of Most persistence: an evolutionary perspective on path dependence and dynamic capabilities, *Organ. Stud.* 32 (2011) 365–382, <https://doi.org/10.1177/0170840610397485>.
- [55] C.M. Beckman, M.D. Burton, Founding the future: path dependence in the evolution of top management teams from founding to IPO, *Organ. Sci.* 19 (2008) 3–24, <https://doi.org/10.1287/orsc.1070.0311>.
- [56] S. Breschi, F. Lissoni, F. Malerba, Knowledge-relatedness in firm technological diversification, *Res. Pol.* 32 (2003) 69–87, [https://doi.org/10.1016/S0048-7333\(02\)00004-5](https://doi.org/10.1016/S0048-7333(02)00004-5).
- [57] F. Belussi, S.R. Sedita, Life cycle vs. multiple path dependency in industrial districts, *Eur. Plan. Stud.* 17 (2009) 505–528, <https://doi.org/10.1080/09654310802682065>.
- [58] E.J. Gómez, R. Atun, Emergence of multilateral proto-institutions in global health and new approaches to governance: analysis using path dependency and institutional theory, *Glob. Health* 9 (2013) 1–17, <https://doi.org/10.1186/1744-8603-9-18>.
- [59] M. He, B. Walheer, Spillovers and path dependencies in the Chinese manufacturing industry: a firm-level analysis, *J. Dev. Stud.* 56 (2020) 817–839, <https://doi.org/10.1080/00220388.2019.1605058>.
- [60] M. Malherbe, F. Simon, Near-field communication (NFC) technology emergence: one or several technological paths? *J. Innovat. Econom. Manag.* 35 (2021) 151–185, <https://doi.org/10.3917/jie.pr1.0091>.
- [61] M. Valorinta, H. Schildt, J.-A. Lamberg, Path dependence of power relations, path-breaking change and technological adaptation, *Ind. Innovat.* 18 (2011) 765–790, <https://doi.org/10.1080/13662716.2011.621745>.
- [62] F. Stache, J. Sydow, Breaking a path by creating a new one: how organizational change boosts children's cancer care, *Organ. Stud.* (2022) 017084062211039, <https://doi.org/10.1177/01708406221103965>.
- [63] A. Abbas, L. Zhang, S.U. Khan, A literature review on the state-of-the-art in patent analysis, *World Pat. Inf.* 37 (2014) 3–13, <https://doi.org/10.1016/j.wpi.2013.12.006>.
- [64] A. Wambganss, L. Tomidei, N. Sick, S. Salomo, E.B. Miled, Machine learning-based method to cluster a converging technology system: the case of printed electronics, *World Pat. Inf.* 78 (2024) 102301, <https://doi.org/10.1016/j.wpi.2024.102301>.
- [65] B. Wang, S. Liu, K. Ding, Z. Liu, J. Xu, Identifying technological topics and institution-topic distribution probability for patent competitive intelligence analysis: a case study in LTE technology, *Scientometrics* 101 (2014) 685–704, <https://doi.org/10.1007/s11192-014-1342-3>.
- [66] S. Wittfoth, Measuring technological patent scope by semantic analysis of patent claims – an indicator for valuating patents, *World Pat. Inf.* 58 (2019) 101906, <https://doi.org/10.1016/j.wpi.2019.101906>.
- [67] E. Mansfield, Patents and innovation: an empirical study, *Manag. Sci.* 32 (1986) 173–181, <https://doi.org/10.1287/mnsc.32.2.173>.
- [68] R.C. Levin, A.K. Klevorick, R.R. Nelson, S.G. Winter, R. Gilbert, Z. Griliches, Appropriating the returns from industrial research and development, *Brookings Pap. Econ. Activ.* 1987 (1987) 783–831, <https://doi.org/10.2307/2534454>.
- [69] R.M. Grant, Strategic planning in a turbulent environment: evidence from the oil majors, *Strateg. Manag. J.* 24 (2003) 491–517, <https://doi.org/10.1002/smj.314>.
- [70] G.C. Unruh, Understanding carbon lock-in, *Energy Policy* 28 (2000) 817–830, [https://doi.org/10.1016/S0301-4215\(00\)00070-7](https://doi.org/10.1016/S0301-4215(00)00070-7).
- [71] P. Berti, L. Levidow, Fuelling expectations: a policy-promise lock-in of UK biofuel policy, *Energy Policy* 66 (2014) 135–143, <https://doi.org/10.1016/j.enpol.2013.09.044>.
- [72] C. Bertram, N. Johnson, G. Luderer, K. Riahi, M. Isaac, J. Eom, Carbon lock-in through capital stock inertia associated with weak near-term climate policies, *Technol. Forecast. Soc. Change* 90 (2015) 62–72, <https://doi.org/10.1016/j.techfore.2013.10.001>.
- [73] P.H. Grünwald, T.T. Cockerill, M. Contestabile, P.J. Pearson, The socio-technical transition of distributed electricity storage into future networks—System value and stakeholder views, *Energy Policy* 50 (2012) 449–457, <https://doi.org/10.1016/j.enpol.2012.07.041>.
- [74] D. Wield, Bioeconomy and the global economy: industrial policies and bio-innovation, *Technol. Anal. Strat. Manag.* 25 (2013) 1209–1221, <https://doi.org/10.1080/09537325.2013.843664>.
- [75] D.F. Oberling, M. Obermaier, A. Szkló, E.L. La Rovere, Investments of oil majors in liquid biofuels: the role of diversification, integration and technological lock-ins, *Biomass Bioenergy* 46 (2012) 270–281, <https://doi.org/10.1016/j.biombioe.2012.08.017>.
- [76] European Commission, Commission Delegated Regulation (EU) 2023/1640 of 5 June 2023 on the Methodology to Determine the Share of Biofuel and Biogas for Transport, Produced from Biomass being Processed with Fossil Fuels in a Common Process, 2023.
- [77] B. Hall, C. Helmers, Innovation and Diffusion of Clean/Green Technology: Can Patent Commons Help? National Bureau of Economic Research, Cambridge, MA, 2011.
- [78] M. Squicciarini, H. Dernis, C. Criscuolo, Measuring Patent Quality: Indicators of Technological and Economic Value, OECD, 2013.
- [79] World Intellectual Property Organization, Handbook on Industrial Property Information and Documentation, WIPO, 1993.
- [80] B. Song, Y. Suh, Identifying convergence fields and technologies for industrial safety: LDA-based network analysis, *Technol. Forecast. Soc. Change* (2018), <https://doi.org/10.1016/j.techfore.2018.08.013>. In Press.
- [81] B. Wang, Z. Wang, Heterogeneity evaluation of China's provincial energy technology based on large-scale technical text data mining, *J. Clean. Prod.* 202 (2018) 946–958, <https://doi.org/10.1016/j.jclepro.2018.07.301>.
- [82] S. Venugopalan, V. Rai, Topic based classification and pattern identification in patents, *Technol. Forecast. Soc. Change* 94 (2015) 236–250, <https://doi.org/10.1016/j.techfore.2014.10.006>.
- [83] M.G. Moehrl, H. Caferoglu, Technological speciation as a source for emerging technologies. Using semantic patent analysis for the case of camera technology, *Technol. Forecast. Soc. Change* (2018), <https://doi.org/10.1016/j.techfore.2018.07.049>. In Press.
- [84] R. Garcia, R. Calantone, A critical look at technological innovation typology and innovativeness terminology: a literature review, *J. Prod. Innovat. Manag.* 19 (2002) 110–132, <https://doi.org/10.1111/1540-5885.1920110>.
- [85] J.M. Utterback, W.J. Abernathy, Dynamic model of process and product innovation, *Omega-Int. J. Manag.* 3 (1975) 639–656, [https://doi.org/10.1016/0305-0483\(75\)90068-7](https://doi.org/10.1016/0305-0483(75)90068-7).
- [86] T. Lager, Product and process development intensity in process industry: a conceptual and empirical analysis of the allocation of company resources for the development of process technology, *Int. J. Innovat. Manag.* 6 (2002) 105–130, <https://doi.org/10.1142/S1363919602000537>.
- [87] M. Bauer, J. Leker, Exploration and exploitation in product and process innovation in the chemical industry, *R D Manag.* 43 (2013) 196–212, <https://doi.org/10.1111/radm.12012>.
- [88] A. Arundel, I. Kabla, What percentage of innovations are patented? Empirical estimates for European firms, *Res. Pol.* 27 (1998) 127–141, [https://doi.org/10.1016/S0048-7333\(98\)00033-X](https://doi.org/10.1016/S0048-7333(98)00033-X).
- [89] C. Stadler, Process innovation and integration in process-oriented settings: the case of the oil industry, *J. Prod. Innovat. Manag.* 28 (2011) 44–62, <https://doi.org/10.1111/j.1540-5885.2011.00860.x>.
- [90] B.D. Barnett, K.B. Clark, Technological newness: an empirical study in the process industries, *J. Eng. Technol. Manag.* 13 (1996) 263–282, [https://doi.org/10.1016/S0923-4748\(96\)01009-0](https://doi.org/10.1016/S0923-4748(96)01009-0).
- [91] S. Wittfoth, M. Wustmans, M.G. Moehrl, The development of product and process claims in blu-ray technology: indicators for the dynamics of innovation theory, in: 2017 Portland International Conference on Management of Engineering and Technology (PICMET), IEEE, 2017.
- [92] S. Wittfoth, T. Berger, M.G. Moehrl, Revisiting the innovation dynamics theory: how effectiveness- and efficiency-oriented process innovations accompany product innovations, *Acad. Manag. Proc.* 112 (2022) 102410, <https://doi.org/10.1016/j.technovation.2021.102410>.
- [93] C. Edquist, L. Hommen, M.D. McKelvey, Innovation and Employment: Process Versus Product Innovation, Edward Elgar Publishing, Cheltenham, U.K., Northampton, Mass, 2001.
- [94] P.L. Robertson, G.L. Casali, D. Jacobson, Managing open incremental process innovation: absorptive capacity and distributed learning, *Res. Pol.* 41 (2012) 822–832, <https://doi.org/10.1016/j.respol.2012.02.008>.
- [95] D.E. Kash, R. Rycroft, Emerging patterns of complex technological innovation, *Technol. Forecast. Soc. Change* 69 (2002) 581–606, [https://doi.org/10.1016/S0040-1625\(01\)00171-8](https://doi.org/10.1016/S0040-1625(01)00171-8).

- [96] Royal Dutch Shell, Annual Report and Form 20-F for the Year Ended December 31, 2017, 2018.
- [97] M. Dijk, R.J. Orsato, R. Kemp, Towards a regime-based typology of market evolution, *Technol. Forecast. Soc. Change* 92 (2015) 276–289, <https://doi.org/10.1016/j.techfore.2014.10.002>.
- [98] European Parliament, Directive 2003/30/EC of the European Parliament and of the Council of 8 May 2003 on the Promotion of the Use of Biofuels or Other Renewable Fuels for Transport, *European Parliament 2003/30/EC*, 2003.
- [99] S. Brown, F. Fai, Strategic resonance between technological and organisational capabilities in the innovation process within firms, *Acad. Manag. Proc.* 26 (2006) 60–75, <https://doi.org/10.1016/j.technovation.2004.08.008>.
- [100] European Commission, Review of the 2012 European Bioeconomy Strategy, 2018. <https://publications.europa.eu/en/publication-detail/-/publication/c2f36c72-2e59-11e8-b5fe-01aa75ed71a1/language-en/format-PDF/source-68980539>. (Accessed 11 July 2018).
- [101] U.S. Department of Energy, THE U.S. the U.S. BIOECONOMY by the Numbers, 2017.
- [102] M. Wustmans, T. Haubold, B. Bruens, Bridging trends and patents: combining different data sources for the evaluation of innovation fields in blockchain technology, *IEEE Trans. Eng. Manag.* (2021) 1–13, <https://doi.org/10.1109/TEM.2020.3043478>.
- [103] C. Block, M. Wustmans, N. Laibach, S. Bröring, Semantic bridging of patents and scientific publications – the case of an emerging sustainability-oriented technology, *Technol. Forecast. Soc. Change* 167 (2021) 120689, <https://doi.org/10.1016/j.techfore.2021.120689>.
- [104] V. Oltra, M. Saint Jean, Sectoral systems of environmental innovation: an application to the French automotive industry, *Technol. Forecast. Soc. Change* 76 (2009) 567–583, <https://doi.org/10.1016/j.techfore.2008.03.025>.
- [105] E. Danneels, The dynamics of product innovation and firm competences, *Strateg. Manag. J.* 23 (2002) 1095–1121, <https://doi.org/10.1002/smj.275>.
- [106] H. Ernst, Patent applications and subsequent changes of performance: evidence from time-series cross-section analyses on the firm level, *Res. Pol.* 30 (2001) 143–157, [https://doi.org/10.1016/S0048-7333\(99\)00098-0](https://doi.org/10.1016/S0048-7333(99)00098-0).
- [107] N. Chatzistamoulou, K. Kounetas, K. Tsekouras, Technological hierarchies and learning: spillovers, complexity, relatedness, and the moderating role of absorptive capacity, *Technol. Forecast. Soc. Change* 183 (2022) 121925, <https://doi.org/10.1016/j.techfore.2022.121925>.

Stefanie Bröring is full professor of Entrepreneurship and innovative Business Models at the Ruhr University Bochum, Germany. Her research focuses on the challenges of technology and innovation management in sustainability transitions as well as the convergence of science, technology and entire industries and need of knowledge integration.

Anna Waßenhoven is a Research Associate (Post-Doc) at the Chair for Entrepreneurship and innovative Business Models at the Ruhr University Bochum, Germany. Her research focuses on strategic technology and innovation management of established companies in transition processes, with an emphasis on strategic renewal.

Silvan Berg is Head of Customer Success at LexisNexis Intellectual Property Solutions. His research focuses on the challenges of emerging technologies and value chains and the transition of technological designs in the setting of the bioeconomy and beyond. He obtained his PhD at the chair for Technology and Innovation Management in Agribusiness at the University of Bonn in Germany.

Michael Rennings, né Wustmans, is an Academic Council at the Ruhr University Bochum in Germany. There, he is working at the Chair for Entrepreneurship and innovative Business Models within the Center for Entrepreneurship, Innovation and Transformation (CEIT). His research focuses on technology dynamics driven by economic transitions and on the impact of transitions on actors' strategic renewal behavior.